SAYAGAIN

NOBODY ON THE CHANNEL NOTICED. THE RADIO LOG DID.

Checks every radio read-back of a safety instruction, and sends the wrong ones to a human.

MISMATCH

H-01 · CH-01 · English

tag: XV-203 ≠ XV-208 XV-208 is on line A (registry)

OPERATOR *H-01-0*

Close valve two-oh-three, line Bravo.

Copy, closing two-oh-eight, line Bravo.

FIELD TECH *H-01-1*

```
zsh · build/
$ jq -r '…' bench/results/text/ladder-r1-reaudit.json
DeepSeek-V4-Flash false-VERIFIED 0 false alarms 4/48
Gemma-4-31B-IT false-VERIFIED 0 false alarms 2/48
MiniCPM5-2B false-VERIFIED 0 false alarms 48/48
Qwen3.8-27B false-VERIFIED 0 false alarms 4/48
regex false-VERIFIED 0 false alarms 0/48
$ jq -r '…' bench/results/text/llm-judge-r1-reaudit.json
llm_judge false-VERIFIED 2 false alarms 1/48
```

CH-01 · HOW THE CHANNEL HEARS IT TODAY

TWO LINES. NOBODY FLAGS THEM.

OPERATOR	Close valve two-oh-three, line Bravo.
FIELD TECH	Copy, closing two-oh-eight, line Bravo.

no flag · no card · the crew walks to XV-208



In the registry the demo runs on, XV-203 is on line B at the press station; XV-208 is on line A, in clarification.

THREE-WAY COMMUNICATION

READ-BACK IS THE CHECK. NOTHING CHECKS IT.

63 % - 50 %

of read-back errors caught: tower, approach

Helmke 2021 · aviation

1 KILLED, 6 INJURED

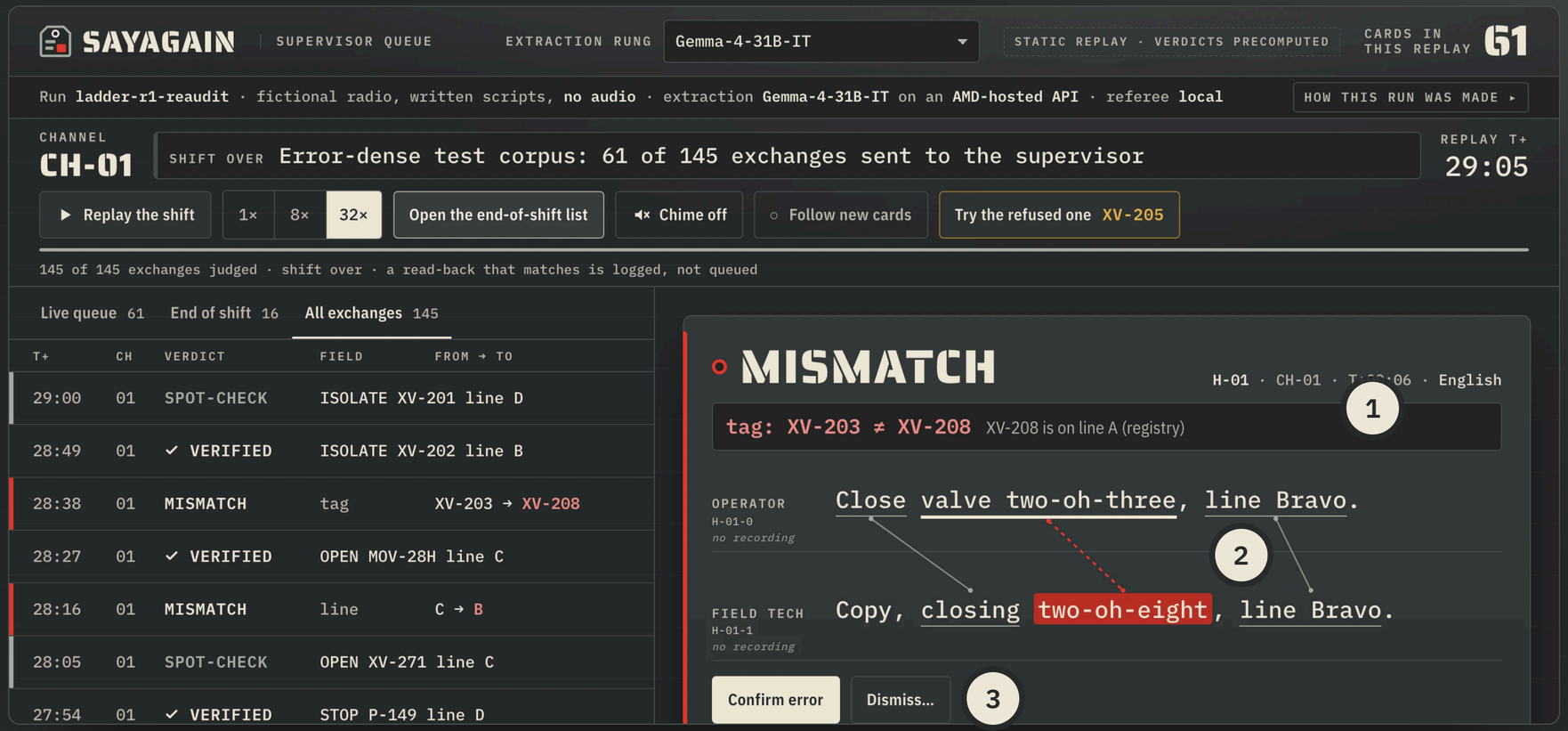
when an isolation valve stayed closed

Goodyear 2008 · CSB

NO RATE

published for plant radio

THE SUPERVISOR GETS THE ONE WRONG WORD.



1. the diff, checked against the registry
2. the read-back word that differs
3. a human confirms or dismisses

ONE MODEL JOB · EVERYTHING ELSE IS CODE

THE MODEL POINTS. CODE DECIDES.

1 · THE READ-BACK, AS THE MODEL SAW IT

Copy, closing two-oh-eight, line Bravo.

H-01-1 · written words, text·scripted

→

2 · WHAT GEMMA-4-31B-IT RETURNED · AMD TOKEN FACTORY


```
{
  "commands": [
    {
      "action_span": "closing",
      "line_span": "line Bravo",
      "state_span": null,
      "tag_span": "two-oh-eight"
    }
  ],
  "kind": "readback",
  "retracted_spans": []
}
```


→

3 · WHAT CODE DECIDED

1. instruction: grounded + covered (1 command(s))
2. read-back: grounded + covered (1 command(s))
3. audio: no gate (no token probabilities, no unprompted pass)
4. instruction: XV-203 line B (registry ok)
5. compare: tag: XV-203 ≠ XV-208

MISMATCH · tag: XV-203 ≠ XV-208

receipt · request c9647a04cfcc4eeaadbfa965a4d74c6d · self-dploy/Gemma-4-31B-IT · developer.amd.com.cn · latency 35.4972 s · cost_usd 4.8937e-06

XV-205 · A VALVE THE PLANT REGISTRY DOES NOT HAVE

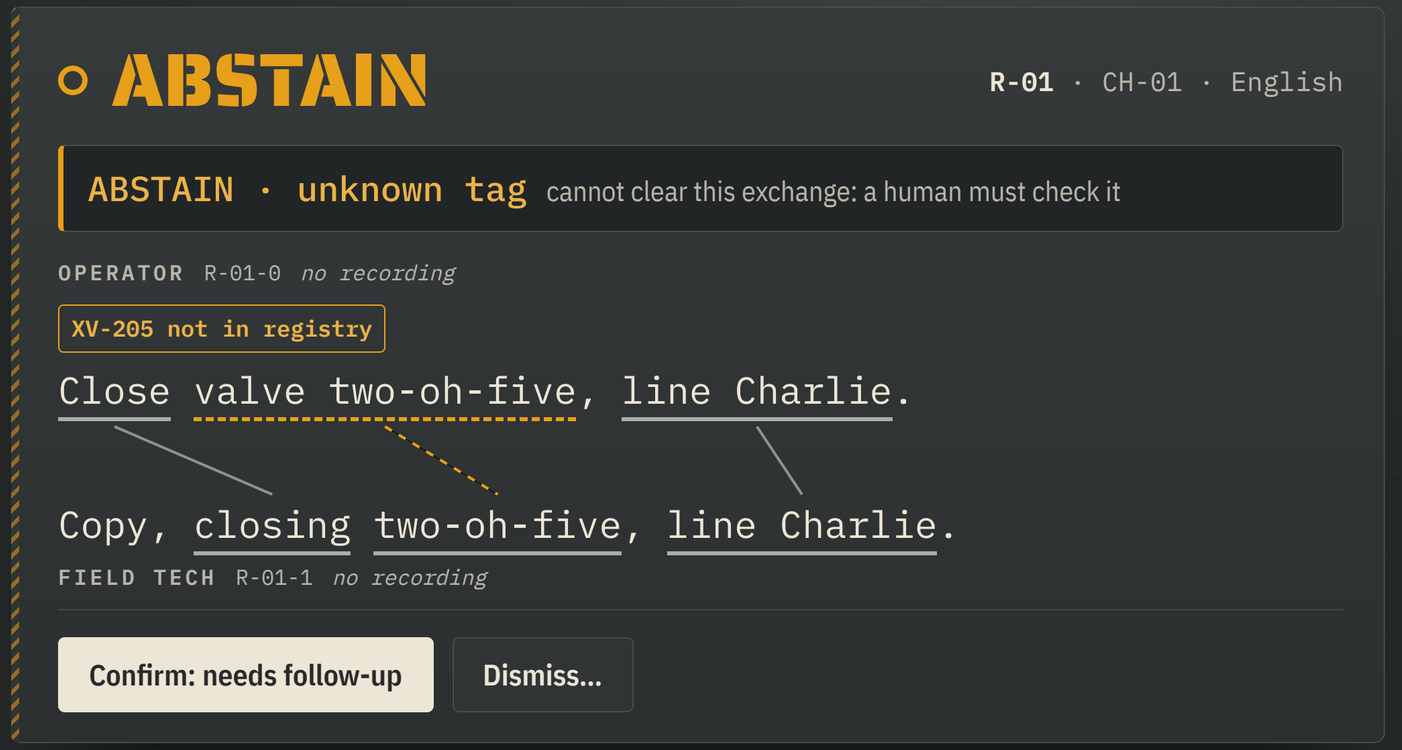
JUDGING IS NOT THE MODEL'S JOB.

LLM-as-judge · DeepSeek-V4.1-Flash, no registry

~~VERIFIED~~

Order and read-back match; the valve does not exist.

It also passed 2 wrong read-backs.



MODEL LADDER · 89 HELD-OUT EXCHANGES · TEXT·SCRIPTED

CLIMB ONLY AS HIGH AS NEEDED.

0 wrong read-backs passed on all five rungs; an LLM judge passed 2.

rung	false-VERIFIED	false alarms on 48 correct read-backs · dashed: budget		rule
regex rung 0 · no model	0		0/48	meets
MiniCPM5-2B smallest model	0		48/48	all 48 to a human
Gemma-4-31B-IT thinking off	0		2/48	meets
DeepSeek-V4-Flash thinking off	0		4/48	over budget
Qwen3.8-27B thinking on	0		4/48	over budget
LLM-as-judge baseline · no registry	2		1/48	passed XV-205

On written words, the regex rung meets the rule too.

STAGE LINE · READ FROM THE RECEIPTS

EVERY STAGE SAYS WHERE IT RAN.

speech-to-text	span extraction	referee	queue
not measured yet	AMD Token Factory · developer.amd.com.cn	local · plain Python	local · FastAPI
Oct 16	a receipt per call	0.77 ms on H-01	reviews in SQLite

11
instruct models probed

8
read the read-back as CLOSE · XV-208 · line B

0
accepted JSON-schema output

0
transcription models

WHO PAYS

PRICED PER CHANNEL. BOUGHT BY HSE.

1,318	mills, Indonesia 2022 (Trase)
× 30 %	addressable*
× 6	channels per mill*
× USD 2,400	per channel-year*
≈ USD 5.7 M	a year, mills only (estimate)

* assumption

WEDGE

free 30-day listen-only baseline

WHO ELSE LISTENS TO PLANT RADIO

LOGGERS TRANSCRIBE. NONE WE CHECKED COMPARES.

	TRANSCRIBES RADIO	COMPARES A READ-BACK	PLANT TAG REGISTRY
NICE Inform AI	**yes**	not in public docs	no
Eventide NexLog + Speech Factor AI	**yes**	not in public docs	not in public docs
Motorola Assist	**yes**	not in public docs	not in public docs
IntelliPERMIT	no	radioed codes, not speech	**yes**
HAAWAII (air-traffic research)	**yes**	**yes, aviation**	no
SayAgain	planned	**yes**	**yes**

Public documentation read 2026-10-09 · sources: evidence/sources.json

README

HONEST BY DESIGN.

> “No recorded speech has been measured.”

> “The plant, its 120 tags and every exchange are fictional…”

> “…what a language model adds is unproven until the audio rows exist.”

NEEDS THE RECORDINGS · PLANNED FOR OCT 16

Audio row
the rule on recorded speech

Failure case
–6 dB look-alike tags

Kill test
English, Bahasa or auto

Blind auditor
by ear vs the queue

THE ASK

PUT IT ON ONE MILL'S RADIO.

ONE MILL. THIRTY DAYS. LISTEN ONLY.

Q1 2027	paid pilot, one palm-oil mill
Q3 2027	DCS check: did it close?
Q4 2027	Malaysian palm belt



github.com/edycutjong/sayagain

EDY CU · JAKARTA